LEHIGH UNIVERSITY

Ishmum Zaman <isz228@lehigh.edu>

Inquiry about Undergraduate Research Assistant Positions

23 messages

Ishmum Zaman <isz228@lehigh.edu> Thu, Apr 23, 2026 at 9:27 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Dear Professor Lee,

I hope you are doing well. I recently came across the undergraduate research assistant openings in Health Data Science & AI/ML Research and Computational Simulation & Network Science Research in Handshake, and I am very interested in applying, as the work here directly goes in the lane that I'm currently building my career upon.

Before I send over my materials, I wanted to ask whether the positions are still open and accepting applications. If it is open, I'll send over all the required materials as soon as possible.

Thank you for your time, and I appreciate any update you can share. I'm looking forward to work under you!

Best,
Ishmum Zaman
Computer Science and Business
Lehigh University

Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Thu, Apr 23, 2026 at 10:58 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Dear Ishmum,

Thank you for your interest. I'm glad to hear the position aligns with your goals.

Yes, I am still considering applications. Please go ahead and send me your **CV and transcript**, and I'll take a look once I receive them.

I look forward to reviewing your materials.

Best regards,
Wayne

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Hsuan-Wei "Wayne" Lee
Assistant Professor, Department of Biostatistics and Health Data Science, Lehigh University
Affiliated Faculty, Department of Mathematics, Lehigh University
Website: https://sites.google.com/view/hsuanweilee/home

Ishmum Zaman <isz228@lehigh.edu> Sun, Apr 26, 2026 at 7:05 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Dear Professor Lee,

Hope your weekend went well. Please find all the attached documents, and let me know if you require any further

information.

I also wanted to add that I'm planning to take Deep Learning (CSE 330) next semester, and if you require it, I'm open to take Reinforcement Learning (CSE 337) or Foundations of Machine Learning (CSE 326) if needed.

Have a great week ahead!
Best,
Ishmum Zaman
Computer Science and Business
Lehigh University
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3 attachments

Resume_Ishmum_Zaman (86).pdf
102K

Statement Of Interest (1).pdf
39K

Unofficial Transcript (4).pdf
274K

Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Mon, Apr 27, 2026 at 12:39 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Dear Ishmum,

Thanks for sending over your materials. I've had a chance to take a look, and I appreciate the effort you put into your application. Your background in software engineering and your experience building ML systems are definitely strong signals.

I think it would be helpful for us to meet and talk through potential directions and expectations, especially to see how we can best structure a collaboration moving forward. Let's find a time to meet sometime in the next two weeks. Please feel free to share a few time slots that work for you, and we'll coordinate from there.

Also, it's great that you're planning to take Deep Learning next semester. That will be very helpful. We can also discuss course planning (including RL or ML) when we meet.

Looking forward to talking more.

Best,
Wayne

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Ishmum Zaman <isz228@lehigh.edu> Tue, Apr 28, 2026 at 10:58 AM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Dear Prof Lee,

I'm free most days after 3pm except for Friday, on Friday I'm free after 4pm. Would you please let me know if this would be like an interview? In that case I will try to prepare and brush up on my ML knowledge. Looking forward to talking to you.

Have a nice week ahead!

Ishmum Zaman
Computer Science and Business
Lehigh University
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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Tue, Apr 28, 2026 at 1:45 PM

To: Ishmum Zaman <isz228@lehigh.edu>

Dear Ishmum,

Thanks for sharing your availability.

How about we talk **this Thursday at 4 PM**? I won't be on campus at that time, so let's meet over video. Here is the link: https://meet.google.com/kue-nizv-fni

Also, this is not an interview, so there's no need to prepare or review ML in advance. It will just be a casual conversation to get to know each other and discuss possible directions. Looking forward to talking with you!

Best,
Wayne

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Ishmum Zaman <isz228@lehigh.edu> Mon, Jun 1, 2026 at 12:53 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Prof Wayne,

Hope you are doing well, just checking in with you. Finished my finals and finally moved to Austin Tx and settled in and have free time to focus on this.

Just wanted to update you, my advisor asked me what the research will be on and to give her a brief outline, and she also asked under what department will this research be done. I saw that you are a professor of College Of Health. I was wondering if it is possible to show this under the CSE department (for example the course number being CSEXXX) because this will make it easier for me to show it as an required elective under CSE department and count towards graduation requirement.

Besides that do you have any draft of what I should be working on that I can share to my advisor or should I make one and run it by you?

Have a great day and looking forward to working under you!

Ishmum Zaman
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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Mon, Jun 1, 2026 at 5:56 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

Great to hear from you, and congratulations on finishing your finals and getting settled in Austin!

Regarding the question about whether this can be listed under the CSE department, I'm honestly not sure. Since the course credit and graduation requirements are handled by your program, I think the best approach is to ask the staff or advisor in your department. They should be able to tell you what options are available and whether an independent study supervised by a faculty member outside CSE can count toward your requirements.

As for the research project, one direction I have been thinking about is the use of large language models for public health decision making. The broad question is whether AI systems can make effective and consistent public health decisions when faced with real-world resource constraints. We would use publicly available public health datasets and construct realistic scenarios involving issues such as substance use and opioid-related harms. The models would be asked to allocate limited resources across different communities while balancing competing goals such as effectiveness, equity, and cost. A key part of the project would be developing a framework to evaluate these decisions systematically. Rather than simply asking whether a model can generate a recommendation, we would study how stable, explainable, and evidence-based those recommendations are. This sits at the intersection of AI, machine learning, decision making, and public health. If the project goes well, I believe it has the potential to lead to a strong journal publication.

Feel free to share this outline with your advisor. Let me know what you think, and we can refine the scope together over the summer.

Best,
Wayne

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Ishmum Zaman <isz228@lehigh.edu> Thu, Jun 4, 2026 at 2:22 AM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Prof Wayne,

This sounds really impressive and realistic. I shared this with my advisor and currently in the process of determining if this can be passed as required elective. I was just curious what kind of tools or languages or frameworks will be involved so that I can watch and learn tutorials and get a head start.

If you have any other info on the project or have any different idea or any info in general please do let me know. I'll keep you updated on what my advisor says.

Have a great week!
Ishmum
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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Thu, Jun 4, 2026 at 1:01 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

Thanks for the thoughtful note, and I am glad the project resonates with you. It is great that you are checking with your advisor about the elective credit. Please keep me posted on what they say, and let me know if it would help for me to send something in writing to support that.

Here is the shape of the project. People are starting to use large language models to help make real decisions, including in public health. We want to study one specific kind of decision, which is how to spread a limited budget across many counties that all have different needs. We will use real public data on opioid overdose deaths, population, and community vulnerability in Pennsylvania counties. The first question is simple to state. When we ask a language model to allocate resources across these counties, what logic does it actually follow, and who gets left behind.

The more interesting part comes next. We expect that models tend to focus on the raw number of deaths in a county and pay less attention to need per person or to how vulnerable a community is. If that is true, smaller and more vulnerable counties could be underserved. We then connect the counties into a network, so that each county agent only sees its own data and messages from its neighbors. The key question is whether that bias gets worse once the agents are linked together, and whether the most connected counties end up capturing more than their fair share. That network angle is what makes the work novel, and it is where most of the new science lives.

Your main role is the engineering and evaluation side, which fits your background well. You will build the data pipeline, set up the agents and the network, and write the evaluation code that measures these biases in a clean and repeatable way. You will also run the same experiments across several different models to check that our findings hold up. I will handle the network design and the statistics, and that is also the part where you will pick up new methods from me as we go. The plan is for this to become a publishable paper, and it also lays the groundwork for a larger project later on.

On tools, the core stack is Python with pandas for the data, LangGraph for running the agents, the OpenAI and Anthropic APIs plus one open model for the language models, networkx for the county network, and statsmodels or scipy for the analysis. None of this is far from what you have already done.

Since you asked about a head start, here are three good first steps you can take at your own pace:

1. Get to know the data. Download county level overdose mortality from CDC WONDER, population and demographics from the American Community Survey, and the CDC Social Vulnerability Index. Try to join them into one clean table of Pennsylvania counties. This alone will teach you a lot about the problem.

2. Build a tiny agent. Using LangGraph and one model API, make a simple version where a single agent reads a few counties of data and returns a budget split. Do not worry about quality yet. The goal is just to get comfortable with the loop.

3. Watch a short intro to networkx (or other network packages you like to use, e.g. igraph) and to basic regression. You do not need depth here. Just enough to recognize what a county network and a simple model fit look like. We will go deeper together.

There is no rush on any of this, especially with your Visa internship running through the summer. Whatever you get to is a bonus. Please feel free to send me emails if you have any questions or progress you want to share. I am happy to talk more whenever you like.

Best,
Wayne
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Ishmum Zaman <isz228@lehigh.edu> Thu, Jun 4, 2026 at 2:55 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Professor Wayne,

I've read through it and love it. I'll go through the resources and look into them. I was planning to learn langchain and langgraph over the summer anyway so this lines up perfectly. I thought at some point of this project we have to train/fine tune models.

I was also hoping to learn PyTorch this summer anyway besides LangGraph. Hence I was wondering if we need to use PyTorch in this project (or if we can incorporate it if needed?). What do you think?

Also I submitted last description you gave me to my advisor and she game me initial approval for this, so it's looking positive! I'll keep you posted.

Thanks and have a great day!

Ishmum Zaman
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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Thu, Jun 4, 2026 at 4:51 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

Great to hear, and thank you for the update! Congratulations on the initial approval from your advisor. Please keep me posted as it firms up.

Good question about training and fine tuning. For the core of this project, the honest answer is that we will mostly use models as they are, through prompting and inference, rather than training them from scratch or fine tuning them. The whole point of the study is to audit how existing models behave when we ask them to allocate resources. So the main results come from running models and measuring what they do, not from building new ones. That said, PyTorch still has a real place here, so your summer plan is not wasted at all. One of our models will be an open model that we run ourselves, and the usual way to do that is through Hugging Face, which sits on top of PyTorch. So the moment you load an open model and get an allocation out of it, you are already inside the PyTorch ecosystem. Learning it now will make that part much smoother for you later.

There is also a natural optional extension where PyTorch becomes central. Once we have measured the bias, a strong follow up question is whether we can reduce it. One way to test that is to lightly fine tune a small open model and see if its allocations become more equitable. That is a real and interesting direction, and it fits your interest perfectly. I would treat it as a stretch goal rather than a requirement, but I am very open to it if the first results look promising.

So here is how I would suggest spending the summer if you want to get ahead. Put LangGraph, the model APIs, and pandas first, since those are on the critical path for the first results. Learn PyTorch and Hugging Face transformers next, since they connect directly through the open model and matter a lot for your longer term goals anyway.

Learn what you enjoy, and we will line it all up when you are back. I am really glad you are excited about this!

Have a great day,
Wayne
[Quoted text hidden]

Ishmum Zaman <isz228@lehigh.edu> Fri, Aug 21, 2026 at 6:21 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Professor Wayne,

Hope you are doing well, it's been a while! I just finished my internship at Visa last week, and am looking forward to coming back to Lehigh this weekend.

First, I wanted to let you know I got the approval. From what I understand, I will be taking CSB 392 under professor Witmer (who is a co-director of CSB) officially. We will work on our project, and whatever resources we use, we have to keep Prof.Witmer in loop, for example, if we use github or google drive, he needs to have access. At the end of the semester, you and professor Witmer will consult and give me an appropriate grade. I've attached the screenshot as reference. Please let me know if you have any questions, or what should be our next course of action!

Have a great week,
Ishmum Zaman

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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Fri, Aug 21, 2026 at 8:40 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

Great to hear from you, and welcome back to Lehigh! I hope you had a great experience at Visa this summer.

This is great news. Thanks for getting everything approved and for explaining how CSB 392 will work. I am happy to keep Professor Witmer in the loop and give him access to the project materials and resources as needed.

Would you like to meet sometime next week? We can catch up, go over where things stand, and map out the first steps and goals for the semester. Let me know what your schedule looks like, and we can find a time that works for both of us.

I'm looking forward to getting started on the project with you!

Best,
Wayne

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Ishmum Zaman <isz228@lehigh.edu> Mon, Aug 24, 2026 at 1:09 AM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Prof Wayne,

I'm pretty much free most days after 4:30 pm starting from Wednesday. Looking forward to meeting you!

-Ishmum
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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Mon, Aug 24, 2026 at 7:31 AM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

Sounds great! Would you like to meet at 5 pm on Wednesday? We can either meet in my office (HST Room 122) or via Zoom.

Looking forward to meeting you!

Best,
Wayne
[Quoted text hidden]

Ishmum Zaman <isz228@lehigh.edu> Mon, Aug 24, 2026 at 12:44 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Prof Wayne,

Yes 5pm Wednesday works great for me. I can come to HST 122. Looking forward to meeting you!

Sincerely,
Ishmum
[Quoted text hidden]

Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Mon, Aug 24, 2026 at 4:50 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Sounds great! See you then!

Best,
Wayne
[Quoted text hidden]

Ishmum Zaman <isz228@lehigh.edu> Wed, Aug 26, 2026 at 2:08 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hey Prof Wayne,

Hope you're doing well. I was wondering if it would be possible to meet over zoom instead of in person meeting? This is because I just moved to a new house and there is some repair going on, and all of my other roommates have classes during time, so someone needs to be in the house. This is a one time issue and we can meet in person any other time.

I apologize if this causes any inconvenience, and I really appreciate your consideration.

Looking forward to hearing from you!

Ishmum
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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Wed, Aug 26, 2026 at 2:58 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

No problem at all! We can meet over Zoom this time. I completely understand. Here is the link:
https://lehigh.zoom.us/j/8387987113

Good luck with the repairs, and I look forward to talking with you soon!

Best,
Wayne

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Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Wed, Aug 26, 2026 at 10:14 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

It's great to talk to you today. Here is the plan for the fall.

The question is whether where a community sits in a network changes how much money it gets, even when its need is identical to someone else's. Our setting is opioid overdose funding in Pennsylvania. Rather than handing one model all 67 counties and asking it to split a budget, we give every community its own language model agent. Each agent sees only its own numbers, plus whatever its neighbors choose to tell it. They talk for a few rounds, each one argues for what it thinks it deserves, and a separate model turns those claims into a final allocation that adds up to the budget. If a community in the middle of the network consistently walks away with more than an identical community out on the edge, that tells us something real about what happens when we wire these models together and let them negotiate.

Proving that is the hard part. Real counties are terrible for this. A county with a lot of neighbors also tends to be a certain size, a certain level of rural, and to carry a certain overdose burden, so you can never tell which one is doing the work. So we randomize. We build clean synthetic networks, a ring, a star, a core and periphery structure, a random graph, and then shuffle the county data across the positions many times over. The same needs land in different spots on every run. Once we do that, any link between centrality and allocation has to come from structure, because we assigned it ourselves. Pennsylvania then shows up at the end as the applied case rather than as the evidence.

One more thing worth knowing. The models already know Philadelphia is big and Cameron County is tiny, so real names leak information we never gave them. Our main runs will use anonymous labels, County A and County B. We will also run a version with the real names, because how much the money moves on names alone is a finding in itself.

Two more layers sit on top of that. The first one you get for free. When the agents talk, some get mentioned and backed up and others get ignored, so the conversation itself is a second network. Parse the message logs, build the network of who paid attention to whom, and we can ask whether the structure we imposed works through the attention the agents actually paid. Hardly anyone does this. Most work in this space reads the final output and throws the conversation away. The second layer is intervention. Once we can measure how big the structural advantage is, we try to shrink it. Change the prompt so agents weigh need per person. Broadcast global information so nobody is arguing blind. Rewire the network so everyone has roughly the same number of neighbors. Showing which of those actually helps is what turns this from an observation into a paper. I am aiming at a Q1 journal, and I think we have a real shot if the pieces hold.

The engineering you already have, so put your reading somewhere else. Start with basic network science, meaning degree, betweenness and eigenvector centrality, what each one actually measures, how they differ, and how to get them out of NetworkX. Then read about why randomized assignment lets you claim causation instead of correlation, since that logic is what the entire design rests on. Get comfortable with regression on log transformed outcomes, because our headline result is a coefficient from a regression of log allocation on log deaths, log population, vulnerability and centrality. You should also know that our observations come in clusters, since many counties share a simulation run, and that this changes how standard errors work. On the domain side, spend a little time on how overdose mortality gets reported. In particular, CDC WONDER hides county death counts below ten, which will bite us in the small counties. Last, read four or five papers on algorithmic fairness in allocation and on multi agent LLM systems. Please send me a short paragraph on each. Not a long summary.

Three things to start on.

1. Build the data table first. One row per Pennsylvania county, with population, overdose deaths, death rate, social vulnerability, poverty and a rural classification. CDC WONDER for the deaths, the American Community Survey for population and economics, the CDC social vulnerability index for vulnerability, and the Census county adjacency file for the network. Pool several years of mortality data. Single year counts in the small counties will mostly be suppressed, and those are exactly the counties we care about. Wherever something is missing or you had to make a judgment call, write it down in the repo. I would rather have an honest note than a clean table I cannot trust.

2. Run a cost pilot in your second week. Five agents, two rounds, nothing fancy. Measure tokens and wall clock time for one run. That number decides the shape of the whole semester, and I want it early, not in November.

3. Then build the smallest thing that runs end to end. Around eight agents, one topology, anonymous labels, a final allocation that adds up to the budget. Save everything. The prompt, the input data, the model name and version, the seed, every message the agents send, the final numbers. Do not tune it. Do not chase a good looking result. If the allocation comes out strange, that is fine and might even be interesting. All I want from this step is a pipeline that gives the same answer twice.

Please feel free to let me know if anything is unclear or if you have any questions. I look forward to collaborating with you!

Best,
Wayne
[Quoted text hidden]

Ishmum Zaman <isz228@lehigh.edu> Fri, Sep 4, 2026 at 9:36 PM
To: Hsuan-Wei Wayne Lee <hsl324@lehigh.edu>

Hi Professor Wayne,

I wanted to send you a quick update on my progress so far and what I plan to work on before we meet next Wednesday.

Over the past week, I worked through the network science concepts you recommended. I reviewed degree, betweenness, and eigenvector centrality, including what each measure captures, how they differ, and how to calculate them using NetworkX. I also went through the basic network structures that will be relevant to the project, particularly ring, star, and random networks, and how these structures create different centrality patterns.

I also spent time on the statistical and experimental-design concepts behind the study. I reviewed the distinction between correlation and causation and, more importantly, why randomly assigning the same community profiles to different network positions allows us to isolate the effect of network structure. I worked through the basics of multiple regression, interpreting coefficients and controlling for need variables, followed by log-transformed outcomes and predictors. I also reviewed why observations within the same simulation run are clustered and why we will need cluster-robust standard errors rather than treating every county-agent observation as independent.

For the literature review, I read five papers across algorithmic fairness in allocation and multi-agent LLM systems. My short summaries are below:

Jo et al. (2023), “Fairness in Contextual Resource Allocation Systems: Metrics and Incompatibility Results.”
Jo et al. develop a framework for evaluating fairness in systems that allocate scarce resources, using homelessness services as a motivating example. They distinguish fairness in resource allocation from fairness in resulting outcomes and show theoretically that these objectives can be incompatible, since equal allocation across groups does not necessarily produce equal outcomes. They also show that vulnerability-based policies can still lead to outcome disparities because vulnerability alone does not capture factors such as treatment effects. This feels relevant to our project because it

emphasizes that we need to define precisely what constitutes a fair allocation based on legitimate need variables before determining whether network position introduces an unjustified allocation advantage.

Li and Varakantham (2022), "Efficient Resource Allocation with Fairness Constraints in Restless Multi-Armed Bandits."
Li and Varakantham study limited-resource allocation in Restless Multi-Armed Bandit systems, including settings motivated by public-health interventions. They introduce fairness constraints intended to prevent individuals or groups from being repeatedly overlooked while the system maximizes overall reward, and propose both planning and learning approaches for satisfying those constraints. Their results show that fairness requirements can be incorporated without necessarily causing a large loss in overall solution quality. This is useful for our project because it provides an example of how public-health resource allocation can involve a tradeoff between efficiency and fairness, although their work does not involve LLM agents or communication networks.

Cookson, Ebadian, and Shah (2026), "Fairness Perceptions of Large Language Models."
Cookson et al. examine how LLMs behave when asked to make fair resource-allocation decisions using synthetic and real allocation settings. They find that simply asking an LLM to make a "fair" allocation does not guarantee that it will follow a particular formal fairness criterion, and the models often favor collective welfare over stronger envy-based fairness notions. They also find that specifying the fairness objective explicitly is more effective than giving a vague instruction to act fairly. This is particularly relevant to our project because our final allocator will also be an LLM, so we need to be careful about how the allocation objective is specified rather than assuming the model has the same definition of fairness that we do.

Zhang et al. (2025), "G-Designer: Architecting Multi-agent Communication Topologies via Graph Neural Networks."
Zhang et al. study how communication topology affects the performance and efficiency of LLM-based multi-agent systems and introduce a method that automatically constructs task-specific communication networks. Their work models agents as nodes in a graph and shows that changing the topology can affect task performance, communication cost, and robustness. This is highly relevant to our study because it supports the idea that the structure connecting LLM agents can materially affect system behavior. The main difference is that their objective is to optimize system-level performance and efficiency, while our study asks whether an individual agent's structural network position creates an allocation advantage independent of actual need.

Shen et al. (2025), "Understanding the Information Propagation Effects of Communication Topologies in LLM-based Multi-Agent Systems."
Shen et al. study how communication topology influences the propagation of correct and incorrect information through LLM multi-agent systems. They use a causal framework to examine this process and find that moderately sparse networks can perform well by allowing useful information to propagate while limiting the spread of errors. This seems especially close to the network component of our work because it directly studies how topology changes information flow between LLM agents. However, their focus is on information propagation and collective task performance rather than whether an individual agent's network position systematically changes the resources it receives.

On the implementation side, I have started building the Pennsylvania county data table. I am currently working through the population, overdose mortality, vulnerability, poverty, rural classification, and adjacency variables and will document any missing values, suppressed mortality counts, or judgment calls rather than silently cleaning them. My goal is to have the table completed before our meeting next Wednesday.

Next week I also plan to begin the cost pilot you suggested after I'm done with the table, preferably before Friday. That should give us an early estimate of how expensive and time-consuming repeated simulations will be before we scale to the larger end-to-end experiment.

Please let me know if there is anything you would like me to prioritize differently before Wednesday.

Best,
Ishmum

[Quoted text hidden]

Hsuan-Wei Wayne Lee <hsl324@lehigh.edu> Sun, Sep 6, 2026 at 5:01 PM
To: Ishmum Zaman <isz228@lehigh.edu>

Hi Ishmum,

Nice update! The reading looks solid and the summaries are the right length. A few things for the week ahead.

Start with the cost pilot rather than the table, and use made up counties for it. Five fake counties with plausible numbers give the same cost estimate as real ones, and the table is the slower job. If you have a number before Wednesday it changes what we can decide in the meeting. While you run it, log more than tokens. Wall clock time, rate limits, failed calls that need a retry, and how often an agent returns something you cannot parse. Also check whether the claims land anywhere near the budget or wildly overshoot it. That tells us early if the protocol needs work.

One design decision to build in now. The thing that converts agent claims into a final allocation should be a plain deterministic rule that scales claims proportionally to hit the budget. If we use a model for that step, we can never separate what the agents did from what the allocator preferred. An LLM allocator can come later as a robustness check.

Your Cookson summary has a consequence worth acting on. If vague fairness instructions do not pin a model to any particular criterion, our baseline prompt should not mention fairness at all. Keep it neutral. The whole point is what the model does when we say nothing. The explicit normative prompt becomes an intervention arm instead. Please write me one sentence on how we differ from G-Designer and Shen. Both ask whether topology changes collective performance. We ask whether an agent's position changes what that agent receives, with need held fixed. That sentence is what the paper rests on. Also search once more for anyone who has looked at individual outcomes by position. Better to know now than in April.

On the table, query WONDER with the years already pooled in a single request instead of pulling single years and summing them. Suppression applies to each displayed cell, so pooling inside the query leaves far fewer counties blocked. Decide up front whether we count all drug overdose deaths or opioid involved deaths only, and record the exact ICD codes from the NCHS definition. If too many counties still drop out, NCHS has model based county level estimates worth checking, though you should confirm the years cover what we need. Do not patch WONDER gaps with a state source. The case definitions differ and the column would end up meaning two things.

Lastly, do not count on seeds for reproducibility. Anthropic's API has no seed parameter, OpenAI's is best effort, and temperature zero is not fully deterministic anywhere. Treat every output as random, repeat runs, and log the model version string, date, temperature, full prompt and raw response. Models get updated quietly and we need to be able to show our numbers came from somewhere.

Please feel free to let me know if you encounter any issues or there are results you want to discuss. Keep going!

Best,
Wayne

[Quoted text hidden]